Bayesian inference of interval censored data with an application to HIV population surveys: A comparison of Hamiltonian Monte Carlo and Metropolis-Hastings sampling algorithms

Research Protocol

In partial fulfilment of the degree

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1. Literature Review

1.1. Introduction and Background

Survival analysis is a branch of statistics focused on modeling time-to-event data, where the outcome of interest is the time until an event occurs, such as disease onset, equipment failure, or patient mortality. Unlike traditional regression models, survival analysis must account for censoring, which occurs when the exact time of event is unknown [1].

Several types of censoring exist and are outlined in Chen et al. [2]. Right censoring is the most common form and occurs when the event of interest has not yet happened by the end of the study period. This can be due to loss to follow-up or withdrawal of the participant from the study. In these cases, the survival time of these subject is known to exceed a certain value but the exact time in unknown. Conversely, left censoring occurs when the event of interest has already occurred before the subject has entered the study but the exact time is unknown. Hence, we only know that the survival time of this subject is less than a certain value. Interval censoring arises when the exact event time is unknown, but it is known to lie between two observation times [3].

Interval censoring might occur in studies with periodic follow-up exams - for example, a patient might produce a positive HIV test during a clinic visit. The exact time of HIV infection is unknown, but what is known is that it happened before the current visit and after the previous visit [4].

1.2. Statistical Foundations and Bayesian Framework

1.2.1. Interval-censored Data: Formal Definitions and Notation

We use the framework proposed by Zhang et al. to provide formal definitions for interval censoring [4]. As mentioned in the previous section, interval censoring occurs when the exact timing of an event, denoted by the random variable T, cannot be precisely observed but is known to lie within a specific interval.

We first define the cumulative distribution function (CDF) of a survival random variable T_i, as $F(t) = P(T \leq t)$ and let the survival function be $S(t) = 1 - F(t)$. Any one of these functions can be determined given that you know the form of the other.

Now, let T_i represent the true event time for the ith subject in a sample of size n. Instead of observing T_i directly, interval-censored data provide observation intervals $\{I_i\}_{i=1}^n$ where $I_i = (L_i, R_i]$. In other words, the event is known to occur after time L_i but no later than time R_i.

Let $\{t_j\}_{j=0}^{m+1}$ denote the unique ordered elements of $\{0, \{L_i\}_{i=1}^n, \{R_i\}_{i=1}^n, \infty\}$ so that $0 = t_0 < t_1 < \cdots < t_m < t_{m+1} = \infty$.

When $L_i = t_0 = 0$, we have the special case of left-censoring, whilst $R_i = t_{m+1} = \infty$ signifies we have a case right-censoring. Standard interval censoring assumes finite intervals with $0 < L_i < R_i < \infty$.

Given these observation intervals, the likelihood function for the interval-censored data under a survival model with parameter vector θ is proportional to:

$$\mathcal{L}(\theta \mid \text{data}) = \prod_{i=1}^{n} \left[S(L_i \mid \theta) - S(R_i \mid \theta) \right] \tag{1}$$

where $S(\cdot)$ denotes the survival function associated with the chosen parametric model. If the interval shrinks into a single point (the case of an exact observation), the likelihood reduces to the probability density function, whereas right-censored observations contribute directly as $S(L_i \mid \theta)$.

1.2.2. Log-Logistic Distribution

The log-logistic distribution has become increasingly popular in survival analysis, especially for modelling non-monotonic survival data that are initially rising to a peak and subsequently declining. This parametric distribution is defined by two parameters, namely, a scale parameter $\lambda > 0$, which is the median of the the distribution, and a shape parameter $\gamma > 0$ [5].

Here we let T represent the survival time following a log-logistic distribution. The probability density function (PDF) of T, for $t,\ \gamma,\ \lambda$ all greater than zero, is known to be

$$f(t \mid \gamma, \lambda) = \frac{\frac{\gamma}{\lambda}\left(\frac{t}{\lambda}\right)^{\gamma-1}}{\left[1 + \left(\frac{t}{\lambda}\right)^{\gamma}\right]^2} \tag{2}$$

The cumulative distribution function (CDF), which describes the probability of experiencing the event before time t, is

$$F(t \mid \gamma, \lambda) = \frac{\left(\frac{t}{\lambda}\right)^{\gamma}}{1 + \left(\frac{t}{\lambda}\right)^{\gamma}}, \quad t > 0 \tag{3}$$

Correspondingly, the survival function, as defined earlier, is expressed as:

$$S(t \mid \gamma, \lambda) = 1 - F(t \mid \gamma, \lambda) = \frac{1}{1 + \left(\frac{t}{\lambda}\right)^{\gamma}}, \quad t > 0 \tag{4}$$

The hazard function can be determined as a ratio of the density function and the survival function. For our log-logistic case we have

$$h(t \mid \gamma, \lambda) = \frac{f(t)}{S(t)} = \frac{\frac{\gamma}{\lambda}\left(\frac{t}{\lambda}\right)^{\gamma-1}}{1 + \left(\frac{t}{\lambda}\right)^{\gamma}}, \quad t > 0 \tag{5}$$

Values of $\gamma < 1$ imply a decreasing hazard over time, $\gamma = 1$ indicates a monotonically decreasing hazard (equivalent to a monotonically decreasing function similar to an exponential distribution), while values of $\gamma > 1$ yield the characteristic inverted-U (non-monotonic) hazard shape.

The analytical convenience of the closed-form solutions for the log-logistic survival and hazard functions significantly simplify parameter estimation, especially within a Bayesian framework. This also reduces computational complexity, enhancing interpretability and practical application.

1.2.3. Bayesian Inference and Posterior Distributions

Under Bayesian methodology, we quantify uncertainty about our model parameters (λ and γ) using their own respective probability distributions. Prior knowledge is the systematically combined with observed data using Bayes' theorem. Formally, Bayes' theorem expresses the posterior distribution of the parameter (vector) $\boldsymbol{\theta}$, given observed data D, as:

$$p(\theta \mid D) = \frac{p(D \mid \theta)\, p(\theta)}{\int p(D \mid \theta)\, p(\theta)\, d\theta} \tag{6}$$

where $p(D \mid \theta)$ denotes the likelihood function reflecting the probability of observing the interval-censored data under the assumed model, and $p(\theta)$ represents our previous knowledge regarding the prior distribution about the parameters [6].

When specifically dealing with interval-censored survival data, the likelihood function $p(D \mid \theta)$ considers the probability that each observed event occurs within the observed intervals $(L_i, R_i]$. Given a parametric survival function $S(t \mid \theta)$, the likelihood for a dataset of n interval-censored observations follows directly from equation (1.1)

$$\begin{aligned} p(D \mid \theta) &= \prod_{i=1}^{n} \left[S(L_i \mid \theta) - S(R_i \mid \theta) \right] \\ &= \prod_{i=1}^{n} \left[\frac{1}{1 + \left(\frac{t_{L_i}}{\lambda}\right)^{\gamma}} - \frac{1}{1 + \left(\frac{t_{R_i}}{\lambda}\right)^{\gamma}} \right] \end{aligned} \tag{7}$$

assuming the observations are independent. Here, $S(\cdot \mid \theta)$ is defined according to the chosen parametric survival model, in this case, the log-logistic distribution.

The next step in Bayesian analysis is specifying prior distributions $p(\theta)$, which can be broadly classified as being either informative or non-informative priors. Informative priors incorporate prior knowledge, expert opinions, or previously obtained results whilst uninformative priors codify a relative lack of knowledge prior regarding our data.

The method by which priors are chosen depends on the distribution of the underlying data. Under certain cases a conjugate prior, which has the same underlying probability distribution

as the posterior, can be chosen which, in turn, greatly simplifies the analytical calculation of the final distribution form of our model. This is, however, an exception and not the rule as posterior distributions are often too complicated to solve analytically. In these cases, computer applications are required to determine the final model form.

For the log-logistic survival model, the parameter vector typically consists of $\boldsymbol{\theta} = (\lambda, \gamma)$, the scale and shape parameters. Under Bayesian inference with interval censoring, the posterior distribution for these parameters is explicitly given as:

$$p(\lambda, \gamma \mid D) \propto p(\lambda, \gamma) \prod_{i=1}^{n} \left[\frac{1}{1 + \left(\frac{L_i}{\lambda}\right)^{\gamma}} - \frac{1}{1 + \left(\frac{R_i}{\lambda}\right)^{\gamma}} \right] \tag{8}$$

where $p(\lambda, \gamma)$ denotes the joint prior distribution for the parameters. We assume that independence is conditional on parameters and weights. Computational methods, such as Markov Chain Monte Carlo (MCMC) algorithms, are typically required to approximate this posterior due to the complexity of analytical solutions in most cases.

1.3. Computational Methods for Bayesian Sampling

Numerous numerical methods have been developed for posterior approximation, each suited to different inferential scenarios. We will focus on methods developed from the Markov chain Monte Carlo (MCMC) algorithms [7].

MCMC algorithms are computational methods used to sample from complex probability distributions, especially in Bayesian analysis. They create a Markov chain that, after a sufficient number of iterations, converges to the target posterior distribution.

In Bayesian sampling, MCMC methods generate a sequence of correlated samples whose distribution approximates the posterior distribution of model parameters. One begins with an initial value, and through iterative steps, the algorithm proposes new values based on a probability mechanism that depends only on the current state (hence the term “Markov chain”).

Among MCMC methods, two prominent algorithms—Metropolis-Hastings (MH) and Hamiltonian Monte Carlo (HMC)—have demonstrated particular utility in challenging posterior sampling scenarios.

1.3.1. Metropolis-Hastings (MH) Algorithm

The Metropolis-Hastings (MH) algorithm is among the most widely used Markov Chain Monte Carlo (MCMC) techniques in Bayesian statistics, particularly for sampling from complex posterior distributions that cannot be directly evaluated or inverted analytically. Originally developed by Metropolis et al. in 1953 and later generalized by Hastings in 1970, the algorithm provides a

systematic mechanism for generating correlated samples from a target distribution $p(\theta \mid D)$, typically a posterior distribution [8, 9].

The core idea underlying the MH algorithm is to construct a Markov chain whose stationary distribution (the *long run* probability distribution a Markov chain settles into) matches the target posterior (the actual distribution of our model). Starting with an initial parameter vector $\theta^{(0)}$, the algorithm iteratively proposes new candidate parameter values and then probabilistically accepts or rejects them based on an acceptance criterion. The general steps can be outlined as follows (Gelman et al., 2013):

1. Initialization:

   Choose an initial value $\theta^{(0)}$ from a suitable starting distribution.

2. Proposal:

   At iteration t, generate a candidate parameter vector $\boldsymbol{\theta}$ from a user-defined proposal distribution $q\left(\theta \mid \theta^{(t-1)}\right)$, which commonly takes the form of a random-walk Gaussian proposal,

   $$\theta^* \sim \mathcal{N}\left(\theta^{(t-1)}, \Sigma\right) \tag{9}$$

   where Σ is a covariance matrix specifying the proposal scale and correlations among parameters.

3. Acceptance Probability:

   Calculate the acceptance probability, α, given by

   $$\alpha = \min\left\{1, \frac{p\left(\theta^* \mid D\right) q\left(\theta^{(t-1)} \mid \theta\right)}{p\left(\theta^{(t-1)} \mid D\right) q\left(\theta \mid \theta^{(t-1)}\right)}\right\} \tag{10}$$

   This step ensures detailed balance, a necessary condition for the Markov chain to have the correct stationary distribution.

4. Acceptance or Rejection:

   Generate a uniform random number $u \sim \mathrm{Uniform}\,(0, 1)$ and accept the proposed candidate as the next state of the Markov chain if $u \leq \alpha$; otherwise, reject it and retain the current state:

   $$\theta^{(t)} = \begin{cases} \theta^*, & \text{if } u \leq \alpha \\ \theta^{(t-1)}, & \text{otherwise} \end{cases} \tag{11}$$

Repeating these steps yields a Markov chain whose samples asymptotically converge to the posterior distribution $p(\theta \mid D)$. Convergence is typically assessed using diagnostic tools such as trace plots, autocorrelation functions, and the Gelman-Rubin statistic ($\hat{R}$) to determine the adequacy of mixing and stationarity of the Markov chain (Brooks & Gelman, 1998).

MH algorithms do, however, face several limitations and challenges. One such challenge is the Random-walk behaviour of MH algorithms. Commonly chosen proposals can be inefficient, leading to high autocorrelation and slow convergence, especially when parameters are correlated, or the posterior distribution is intricate. There are alternative approaches, like Hamiltonian Monte Carlo, which directly address many limitations.

1.3.2. Hamiltonian Monte Carlo (HMC)

Hamiltonian Monte Carlo (HMC) is an advanced Markov Chain Monte Carlo algorithm specifically designed to overcome many of the inefficiencies encountered with standard Metropolis-Hastings approaches, particularly when sampling from complex, high-dimensional posterior distributions. Initially proposed by Duane et al. (1987) and extensively developed by Neal, HMC exploits gradient information derived from the posterior distribution, incorporating principles from physics of Hamiltonian mechanics to efficiently explore parameter spaces with complex geometries [10, 11], .

HMC improves sampling by adding extra "momentum" variables, allowing the process to mimic physical movement. This method transforms the sampling problem into one of simulating a particle moving under physical forces, where the particle's total energy guides its path. By incorporating gradient information of the target distribution, HMC can navigate complex probability landscapes more efficiently.

In simpler terms, HMC works by introducing extra variables—called momentum—that pair with the original parameters θ. Instead of sampling directly from the target posterior $p(\theta \mid D)$, HMC creates a joint distribution for both θ and the momentum p. Typically, the momentum p is assumed to follow a normal distribution and is independent of θ. The combined distribution is written as:

$$p(\theta, p \mid D) \propto \exp\{-H(\theta, p)\} \tag{12}$$

where $H(\theta, p)$ is the total energy (Hamiltonian) of the system. This energy is split into two parts:

- Potential energy: $U(\theta) = -\log p(\theta \mid D)$, which reflects how the target posterior behaves.
- Kinetic energy: $K(p) = \frac{1}{2} p^T M^{-1} p$, where M is a mass matrix (often chosen to be diagonal).

The key idea is that the gradients (derivatives) of the potential energy, $\nabla_\theta U(\theta)$, provide information about the "slope" of the probability landscape. This helps the sampler move in directions that are more likely to yield valid samples, allowing HMC to explore the parameter space more effectively than methods that do not use gradient information.

Algorithmic Details HMC generates posterior samples through deterministic Hamiltonian dynamics interspersed with stochastic momentum resampling. The standard HMC algorithm

proceeds as follows:

1. Initialization:

   Start with an initial parameter vector $\theta^{(0)}$.

2. Momentum Sampling:

   Sample a new momentum vector $p^{(t)} \sim \mathcal{N}(0, M)$.

3. Leapfrog Integration:

   The algorithm then simulates a trajectory using Hamiltonian equations of motion, discretized by a symplectic integrator known as the leapfrog integrator. Given step size ϵ and number of leapfrog steps L, the leapfrog steps are iteratively computed as:

$$\begin{aligned} p &\leftarrow p - \frac{\epsilon}{2}\nabla_\theta U(\theta), \\ \theta &\leftarrow \theta + \epsilon M^{-1} p, \\ p &\leftarrow p - \frac{\epsilon}{2}\nabla_\theta U(\theta) \end{aligned} \tag{13}$$

   These updates are repeated L times to propose a new state (θ, p).

4. Metropolis Acceptance Step:

   The proposed state (θ, p) is accepted with probability:

$$\alpha = \min\left[1, \exp\{-H(\theta, p) + H\left(\theta^{(t-1)}, p^{(t)}\right)\}\right] \tag{14}$$

   If rejected, the chain remains at the current state $\theta^{(t-1)}$.

The performance of HMC critically depends on the careful tuning of leapfrog step size ϵ, number of leapfrog steps L, and the mass matrix M.

The primary advantage of Hamiltonian Monte Carlo (HMC) is its efficient exploration of posterior distributions, achieved by leveraging gradient information to systematically guide sampling through high-probability regions. This approach significantly reduces autocorrelation between samples, resulting in higher effective sample sizes per iteration compared to traditional random-walk Metropolis-Hastings methods [11]. Additionally, HMC effectively handles correlations and complexities inherent in high-dimensional posterior spaces, making it particularly advantageous for hierarchical or multilevel models commonly encountered in survival analysis.

1.3.3. Existing Studies

Recent literature has extensively compared the performance of Metropolis–Hastings (MH) and Hamiltonian Monte Carlo (HMC) methods in various contexts. Foundational works such as Gelman et al., Brooks, and Hoffman and Gelman provide a theoretical framework that illustrates

how HMC leverages gradient information to explore high-dimensional parameter spaces more efficiently than the traditional MH algorithm, which often relies on random-walk proposals [6, 12, 13]. Empirical studies further support these insights, demonstrating that HMC achieves higher acceptance rates, shorter autocorrelation lengths, and faster convergence in complex models. For instance, studies in reliability engineering and fault model estimation have shown that while MH is relatively straightforward to implement, it tends to underperform in high-dimensional or tightly constrained scenarios compared to HMC, which adapts its trajectory according to the geometry of the target posterior [14, 15].

In the specific realm of interval-censored Bayesian analyses, the literature reveals a notable gap in direct comparisons between MH and HMC. Several studies have applied MH-based methods to interval-censored data, particularly in survival analysis and reliability contexts, as exemplified by work on Weibull, gamma, and log-normal models [16]. Although these studies demonstrate the utility of MH for handling censoring mechanisms, there is a growing recognition that the improved sampling efficiency of HMC could offer significant advantages in these applications. The ability of HMC to effectively navigate complex posterior landscapes suggests that it may yield better mixing and convergence properties in the presence of interval censoring. However, direct empirical evaluations contrasting MH and HMC in this specific context remain limited. This gap highlights an opportunity for further research to determine under which conditions HMC outperforms MH in interval-censored survival analyses, thereby providing clearer methodological guidance for practitioners working with censored lifetime data.

2. Problem Statement

Accurate analysis of interval-censored time-to-event data is an important tool in for any statistician working with healthcare data, where events are often only known to have occurred between specific time-points (assuming they happened at all). The log-logistic model is often chosen for such data because its inverted-U hazard accommodates non-monotonic risk. Fitting this model within a Bayesian framework remains computationally demanding: the likelihood must integrate over each censoring interval, the posterior is typically non-conjugate, and conventional Metropolis-Hastings (MH) samplers tend to mix slowly. In parallel, complex surveys introduce sampling weights that further distort the posterior geometry, compounding the inefficiency of random-walk algorithms.

Hamiltonian Monte Carlo (HMC) offers a principled alternative by using first-order gradients to follow high-probability trajectories through the parameter space, thereby reducing autocorrelation and accelerating convergence. While HMC has proven superior to MH in several high-dimensional settings, no systematic evaluation exists for weighted, interval-censored survival models—precisely the scenario encountered in many large-scale health surveys. Conse-

quently, practitioners lack evidence-based guidance on whether the additional implementation effort of HMC is justified when analysing realistically complex survival data.

3. Rationale

The literature review shows three converging threads:

1. Population-based surveys which incorporate biomarker testing, such as HIV testing, lead to interval-censored outcomes that violate assumptions of traditional right-censoring methods.

2. Its closed-form survival and hazard functions simplify likelihood construction for data with non-monotonic risk, making it a pragmatic modelling choice in health related research.

3. Empirical studies outside the complex survey design, interval-censoring domain report higher effective sample sizes, better exploration of multimodal distributions, and shorter computation times for HMC. However, these advantages have not been demonstrated for interval-censored data, particularly in the presence of survey weights that induce additional curvature in the posterior [17].

Despite its relevance to wide areas of medical research, there remains an absence of head-to-head comparisons between MH (and related Gibbs schemes) and HMC for weighted, interval-censored log-logistic regression. Researchers are therefore forced to choose algorithms based on anecdotal experience rather than systematic evidence.

By simulating realistic weighted interval-censored datasets, fitting Bayesian log-logistic models via MH and HMC, and benchmarking their efficiency on both synthetic and real survey data, this study will contribute the first comprehensive evidence on how—and how well—HMC tackles the challenges of weighted interval-censored survival analysis. The findings will equip researchers with concrete recommendations and promote more reliable inference in fields where time-to-event outcomes drive critical decisions.

4. Research Question

In Bayesian log-logistic regression for interval-censored survival data that incorporate complex survey weights, does Hamiltonian Monte Carlo (HMC) yield demonstrably greater sampling efficiency and posterior accuracy than standard Metropolis-Hastings MCMC, when estimating the effects of key covariates?

5. Hypothesis

Null hypothesis ($\mathbf{H}_0$): There is no difference in sampling efficiency—measured by effective sample size per second (ESS/s), convergence diagnostics ($\hat{R}$), bias, and computation time—between Hamiltonian Monte Carlo (HMC) and Metropolis–Hastings/Gibbs (MH) algorithms when fitting Bayesian log-logistic models to weighted interval-censored survival data.

Alternative hypothesis ($\mathbf{H}_1$): Hamiltonian Monte Carlo (HMC) yields significantly greater sampling efficiency—higher ESS/s, faster convergence ($\hat{R}$ closer to 1 in fewer iterations), lower Monte Carlo error, and/or reduced computation time—than Metropolis–Hastings/Gibbs (MH) for Bayesian log-logistic regression with weighted interval-censored data.

6. Aim

To systematically determine whether Hamiltonian Monte Carlo provides superior sampling efficiency and inferential accuracy, relative to conventional Metropolis–Hastings MCMC, when estimating Bayesian log-logistic regression models for weighted interval-censored survival data generated through simulation and observed in complex survey settings.

7. Objectives

1. Develop a simulation framework that generates interval-censored survival data with non-monotonic hazards and embedded survey weights, reflecting realistic epidemiological survey designs.

2. Implement Bayesian log-logistic regression models for these data using two sampling paradigms—

   (a) Metropolis–Hastings/Gibbs; and

   (b) Hamiltonian Monte Carlo

3. Quantitatively compare sampler performance by evaluating effective sample size per second, convergence diagnostics ($\hat{R}$, autocorrelation), bias, root-mean-squared error, and computational cost across varying censoring severities, weight dispersions, and covariate dimensionalities.

4. Apply both samplers to a complex survey dataset with interval-censored outcomes and design weights, assessing agreement in posterior summaries, credible-interval widths, and practical run-time metrics.

8. Methodology

8.1. Study Design

This study employs a methodological comparative design structured into two complementary phases: a simulation-based experimental component and a secondary analysis of real-world survey data.

8.1.1. Simulation Study

The first phase is a controlled Monte Carlo simulation experiment designed to generate interval-censored datasets under known conditions. After setting a random-seed for reproducibility, event times are drawn from a log-logistic distribution with a non-monotonic hazard function, and covariate effects are embedded through a log-linear model on the scale parameter. Simulated datasets incorporate:

- Binary and continuous covariates;
- Varying levels of censoring severity (20%, 40%, and 60%), sample size, and weight dispersion;
- Survey weights sampled from a Gamma distribution and normalized to reflect complex sampling designs.

The simulation allows for objective benchmarking of both algorithms on metrics such as effective sample size per second (ESS/s), bias, root mean squared error (RMSE), Monte Carlo standard error (MCSE), and convergence diagnostics (e.g., $\hat{R}$).

8.1.2. Empirical Survey Analysis

The second phase involves the application of both sampling algorithms to a real-world dataset: the Zimbabwe Population-Based HIV Impact Assessment (ZIMPHIA) survey. This phase constitutes an observational analytical study using secondary data collected under a complex survey design. ZIMPHIA is a nationally representative survey that includes biomarker-based HIV testing data collected at periodic intervals, making it well-suited for interval-censored survival analysis.

Participants' event times of HIV infection are not observed directly but are known to occur between two points: time at first sexual intercourse and the date of they survey HIV test, thereby forming natural interval-censored survival data. In addition, ZIMPHIA employs multi-stage stratified cluster sampling, with each respondent assigned a survey weight that reflects unequal selection probabilities and non-response adjustments.

The Bayesian log-logistic regression model is fitted to this data using both Metropolis–Hastings/Gibbs and Hamiltonian Monte Carlo algorithms. Sampling weights are incorporated through a pseudo-likelihood formulation, and the models adjust for relevant covariates such as age, sex, and behavioural risk factors.

Comparative evaluation will focus on posterior summaries, convergence diagnostics, effective sample sizes, and computational efficiency, assessing whether the algorithmic advantages observed in the simulation phase generalize to the complexities of real-world population-based data.

8.1.3. Integration of Phases

The two phases are methodologically aligned in terms of model specification, performance metrics, and computational implementation. Findings from the simulation guide the interpretation of empirical results, while insights from the empirical analysis may inform refinements in the simulation (e.g., inclusion of frailty or additional covariates).

This dual-phase design ensures that both internal validity (through controlled simulation) and external validity (through real-data application) are addressed, enabling a robust assessment of whether HMC offers practical improvements over MH in complex Bayesian survival analysis settings.

8.2. Study setting

The empirical phase of this study uses secondary data obtained from the Zimbabwe Population-Based HIV Impact Assessment (ZIMPHIA), is a nationally representative, cross-sectional survey designed to measure the prevalence and impact of HIV in Zimbabwe [18]. It is part of the broader PHIA Project (Population-based HIV Impact Assessment), supported by organizations such as the U.S. Centers for Disease Control and Prevention (CDC), ICAP at Columbia University, and Zimbabwean health authorities. The ZIMPHIA survey to be used in this study was carried out between 2019 and 2020 across all 10 provinces of Zimbabwe and was designed to estimate HIV prevalence, incidence, and treatment coverage in the general population.

The survey employed a multi-stage stratified cluster sampling design and included biomarker testing, structured interviews, and household demographic data. The data are de-identified and publicly available under data use agreements, with appropriate ethical approval obtained by the main website of the study.

8.3. Study population

8.3.1. Sample characteristics

Out of 11,707 eligible households, 89.1% completed the household interview. Among 22,751 eligible adults aged 15 years and older (13,290 women and 9,461 men), 19,535 individuals (11,871 women and 7,664 men) completed both the interview and HIV biomarker testing, yielding an overall adult response rate of 76.5% (79.6% for women, 72.2% for men).

HIV prevalence among adults was 12.9%, with higher rates among women (15.3%) compared to men (10.2%). Annual HIV incidence was estimated at 0.38%, corresponding to approximately 31,000 new cases per year, and viral load suppression among all HIV-positive adults was 77.3%. These characteristics provide sufficient demographic and clinical variability for evaluating covariate effects in the proposed Bayesian log-logistic regression models.

8.3.2. Inclusion criteria

Participants will be included in the analysis if they meet all of the following criteria:

1. Aged 15 years or older at the time of interview.
2. Completed the individual questionnaire.
3. Possess complete sampling information, including individual weights, stratum, and PSU identifiers.
4. Agreed to be tested for HIV as part of the survey.
5. Provided written informed consent (or assent with guardian consent for minors aged 15–17), as required under the original ZIMPHIA ethical approvals.

Participants with missing key variables (weights, strata, test dates) will be excluded from the main analysis but may be included in sensitivity analyses if relevant.

8.3.3. Sample size

The analytic sample for this study will be drawn from the 19,535 adults who completed both the interview and biomarker testing. However, since interval-censored survival models require information on the timing of seroconversion events, only participants with at least two documented HIV testing encounters will be eligible for inclusion in the survival analysis. The final sample size will therefore be a subset of the full ZIMPHIA cohort and is expected to remain sufficiently large ($n > 10,000$) to support stable Bayesian estimation and meaningful comparisons between sampling algorithms.

In parallel, the simulation study in Phase I will mimic this empirical sample size and weight structure to allow for generalisable algorithm performance comparisons under conditions

reflecting the ZIMPHIA design. This design ensures that computational efficiency and convergence properties of Hamiltonian Monte Carlo (HMC) and Metropolis–Hastings (MH) algorithms can be evaluated under realistic data constraints.

8.4. Data analysis

The analysis is structured into two complementary parts: firstly a simulation study designed to systematically assess the performance of Bayesian sampling algorithms under controlled conditions. Secondly an empirical analysis applying these methods to real-world HIV incidence data from the ZIMPHIA 2020 survey.

Analysis will done using an Apple MacBook M2 Pro using R statistical software, R version 4.4.1 (2024-06-14) [19].

8.4.1. Simulation Study Analysis

Data Generation

For each simulation scenario, synthetic datasets are generated using a parametric log-logistic accelerated failure-time (AFT) model. Specifically, we simulate interval-censored survival data reflecting the context of HIV infection, as follows:

- Covariate Generation:
  - Sex (X_1): Simulated from a Bernoulli distribution with probability p = 0.55, reflecting the sex distribution of adult respondents in ZIMPHIA.
  - Age (X_2): Drawn from a truncated normal distribution $\mathcal{N}(30,\ 10^2)$ with truncation at ages 15 and 65 years.
- Interval-Censoring Framework: Participants' interval-censoring bounds are defined by:
  - Lower Bound (L_i): Age at first sexual intercourse, simulated as $A_{0i} \sim \text{Uniform}(12, X_{2i})$.
  - Upper Bound (R_i): Calculated as $A_{1i} = A_{0i} + T_i$, where T_i represents the true, latent time to HIV seroconversion.
- Survival Time Simulation: True survival times (T_i) are simulated from the log-logistic distribution with individual-level scale parameter: $\log(\lambda_i) = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i}$, and shape parameter γ.
- Parameters are set as follows:
  - Baseline intercept: $\beta_0 = \log(5)$ (baseline median survival = 5 years).
  - Covariate effects: $\beta_1 = -0.5$ (sex), $\beta_2 = -0.2$ (age).

  - Shape parameter: $\gamma = 2.0$, indicating a non-monotonic hazard function (initially increasing and then decreasing over time).

- The latent event times T_i are generated using the inverse cumulative distribution function (CDF) method:
  - $T_i = \lambda_i \left(\frac{U_i}{1-U_i}\right)^{\frac{1}{\gamma}}$, where $U_i \sim \mathrm{Uniform}(0, 1)$.

  Individuals whose simulated age at event (A_{1i}) exceeds the maximum follow-up age of 65 years are treated as right-censored.

- Survey Weights Simulation: Sampling weights are simulated from a Gamma distribution to emulate complex survey designs:
  - Low dispersion scenario: $Gamma(k = 10, \theta = 0.1)$, reflecting minimal variation.
  - High dispersion scenario: $Gamma(k = 1, \theta = 1)$, representing high variability in weights.

All weights are normalized to sum to the sample size (n) within each simulated dataset.

Model Estimation and Evaluation

For each generated dataset, Bayesian log-logistic models are fitted using two sampling algorithms:

- Metropolis–Hastings/Gibbs (MH):

Four parallel chains per replication, each with 1,000 burn-in iterations followed by 5,000 sampling iterations. Gaussian random-walk proposals are tuned to achieve approximately 20–40% acceptance rates.

- Hamiltonian Monte Carlo (HMC):

Four parallel chains per replication, each comprising 1,000 warm-up and 1,000 sampling iterations, using the No-U-Turn Sampler (NUTS). The default adaptation settings in Stan are initially employed, with adjustments made if divergences occur.

Performance metrics collected for each algorithm include:

- Effective sample size per second (ESS/s), representing computational efficiency.
- Convergence diagnostics, specifically the Gelman–Rubin statistic ($\hat{R}$), with convergence defined as $\hat{R} < 1.1$.
- Posterior accuracy and precision, assessed through bias, root-mean-squared error (RMSE), and credible-interval width.
- Computational resources, recorded as total CPU runtime and memory usage.

Results are aggregated across 500 replications per scenario, and comparative analyses between MH and HMC are summarized through relative efficiency metrics and statistical tests (e.g., Wilcoxon signed-rank tests).

8.4.2. Empirical ZIMPHIA Data Analysis

Data Preparation

The empirical analysis uses secondary data from the ZIMPHIA 2020 survey. Interval-censored survival data are constructed by defining each individual's interval as follows:

- Lower bound: Age at first sexual intercourse (reported by respondent).
- Upper bound: Age at HIV-positive diagnosis obtained during survey testing; individuals who test negative at survey time are right-censored at their current age, with a maximum censoring age set at 65 years.

The ZIMPHIA complex survey design, including stratification, clustering, and sampling weights, is explicitly incorporated into the analysis. Sampling weights provided by ZIMPHIA are normalized and integrated into the Bayesian log-logistic likelihood via a weighted pseudo-likelihood approach.

Model Specification and Estimation

We fit the Bayesian log-logistic AFT survival model to ZIMPHIA data using both MH and HMC methods, with identical prior distributions and sampling specifications as outlined in the simulation study analysis.

Evaluation of Algorithmic Performance

The comparative performance of MH and HMC on the ZIMPHIA dataset is assessed through the following measures:

- Convergence diagnostics: Gelman–Rubin statistics and effective sample sizes to evaluate algorithm mixing and stability.
- Computational efficiency: Effective sample size per second (ESS/s) and total runtime.
- Posterior inference consistency: Comparison of posterior distributions for key covariates (sex and age), specifically examining posterior medians and credible intervals.

Sensitivity Analysis

To assess robustness, sensitivity analyses are performed:

- Weight trimming: Extreme sampling weights exceeding the 99th percentile are trimmed to assess their influence on posterior estimates and convergence behavior.

- Prior specification adjustments: Priors on regression parameters and the shape parameter are varied to evaluate the sensitivity of posterior estimates to prior choices.

Reporting

Results from the empirical analysis are presented through:

- Tables comparing covariate effect estimates and associated uncertainty (credible intervals) between MH and HMC methods.
- Visual summaries (e.g., bar charts, boxplots) highlighting differences in computational efficiency (ESS/s, convergence diagnostics).
- Narrative interpretation of substantive findings (effects of sex and age on HIV risk) and implications for practical implementation of Bayesian sampling methods.

9. Ethics and regulatory compliance

9.1. Approval by regulatory authorities

Prior to commencement, ethical approval will be obtained from the Health Research Ethics Committee (HREC) of Stellenbosch University, which is registered with the National Health Research Ethics Council (NHREC) of South Africa.

For the secondary data analysis component, access to and use of the ZIMPHIA 2020 survey dataset will be subject to approval by the Population-based HIV Impact Assessment (PHIA) Project. A formal data use agreement will be requested and adhered to, and the researcher will comply with all applicable conditions for the ethical handling and analysis of de-identified individual-level survey data.

The study will be implemented with full respect for the rights, dignity, and welfare of participants, and no primary data collection involving human participants will be undertaken. As such, the project involves minimal risk and qualifies as a secondary data analysis of anonymised public-use data, subject to appropriate approvals.

9.2. Social value

This research addresses a methodological gap in the application of Bayesian methods for complex survival data, specifically interval-censored outcomes in the context of weighted national health surveys.

The primary beneficiaries of this research include:

- Biostatisticians and epidemiologists, who increasingly rely on Bayesian frameworks for analysing survival and longitudinal data from large surveys with complex sampling

designs.

- Health researchers and policymakers working with HIV surveillance data, for whom improved analytical methods can lead to more reliable and timely estimates of disease incidence and risk factors.
- Data analysts and methodologists in national and international health agencies (e.g., ministries of health, UNAIDS, CDC) who require robust, reproducible Bayesian models to inform public health strategies.

At a broader level, by promoting methodological rigour and computational efficiency in the analysis of HIV-related health outcomes, this study supports evidence-based decision-making in public health, particularly in high-burden countries. The findings are expected to facilitate more accurate and efficient secondary analysis of complex survey data, ultimately contributing to better understanding, monitoring, and control of the HIV epidemic in Southern Africa and similar settings globally.

10. Study limitations and assumptions

10.1. Limitations

Several limitations inherent to the design and scope of this study must be acknowledged:

- Generalizability:
  The simulation scenarios, while systematically varied, are still constructed under specific parametric assumptions (e.g., log-logistic survival distribution, particular covariate effects, and weight structures). Consequently, findings regarding the relative performance of Hamiltonian Monte Carlo (HMC) and Metropolis–Hastings/Gibbs (MH) algorithms may not generalize to all survival models, particularly those with different censoring mechanisms, non-parametric structures, or extreme posterior geometries.
- Simplified Covariate Structure in Simulation:
  The simulation study includes only two covariates (one binary, one continuous) and does not model interactions, time-dependent covariates, or more complex missing data patterns. Real-world data often exhibit greater complexity, which may affect sampler performance differently.
- Application Limited to ZIMPHIA 2020:
  The empirical validation is based solely on the ZIMPHIA 2020 dataset. While this dataset is nationally representative for Zimbabwe, findings from the empirical analysis may not be fully transferable to other contexts, populations, or health conditions with different sampling designs or epidemic profiles.

10.2. Assumptions

The study rests on several key assumptions:

- The log-logistic model is an appropriate data-generating mechanism for the simulated survival times.
- Survey weights used in both simulation and empirical analysis appropriately adjust for selection bias and representativeness, and their normalization does not introduce significant bias into posterior estimates.
- We assume that weights are fixed.
- Prior distributions specified for model parameters are sufficiently weakly informative to allow the data to dominate the posterior without unduly influencing convergence behavior.
- The performance of the sampling algorithms (MH and HMC) can be fairly compared across methods, assuming that tuning (e.g., proposal variance for MH, step-size for HMC) is reasonably optimized within practical limits.

11. Conflict of interest

The principal investigator declares that there are no actual or potential conflicts of interest related to the design, execution, or dissemination of this study.

12. Study timeline

Figure 1 represents the proposed project timeline for the synthesis and submission of the research project.

13. Study budget

There are no direct costs associated with the implementation of this study, as it involves the secondary analysis of publicly available survey data and the simulation of synthetic datasets using existing computational resources.

The principal investigator is supported through an NIH D43 training grant, which provides funding for research-related activities, including support for the dissemination of findings through scientific publications and conference presentations.

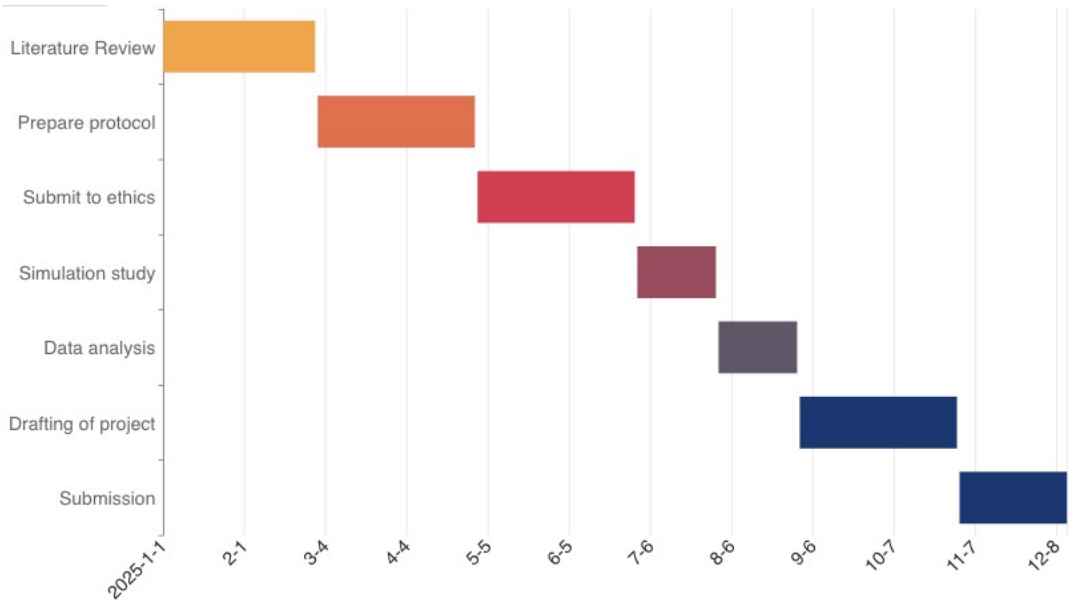

Figure 1: Gantt chart of proposed project timelines

14. Dissemination of results

The findings of this study will be shared with both academic and applied stakeholders through multiple dissemination channels. The results will be submitted for publication in a peer-reviewed open-access journal in the fields of biostatistics or epidemiological methodology. The research project will adhere to the STROBE (Strengthening the Reporting of Observational Studies in Epidemiology) guidelines, as recommended by the Equator Network for studies involving observational or secondary data.

The study will be presented at the Annual Conference of the South African Statistical Association (SASA), providing an opportunity for peer feedback and methodological discussion within the national statistical community.

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